

ITT-307 - BYOD Deployment Plan

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BYOD and IoT devices provide organizations and their users with ease of access, convenience, and efficiency. Unfortunately, these technologies can also provide a lot of security vulnerabilities when organizations fail to configure and implement security tools/management systems on those devices. Some of these security challenges include but are not limited to physical vulnerabilities if the device is stolen or misplaced, limited updates depending on the device, connectivity vulnerabilities, and accessing untrusted content.

Physical security is essential because if users use to bring their own devices or utilize any Internet of things devices the risk of access to sensitive information or to company systems is increasingly higher when the physical technology ends up in the hands of the threat actor. Though BYOD boosts user morale because of convenience, depending on the device, OTA updates are essential but are not always set up to occur automatically long term. When users or the organization fail to ensure updates are being made to their firmware, patches, and important solutions to vulnerabilities can be taken advantage of by actors. Connection vulnerabilities whether it is through a USB flash drive or USB cable, hotspot, tethering, etc. can be exploited by a threat actor. Depending on the management of the specification on what users can do on their devices, there is risk in allowing them to go around limitations and download third-party applications that may contain vulnerabilities or even malicious code.

The main mobile devices that will be utilized by users within the organizations are tablets, smartphones, and laptops, each with MDM implemented. Every mobile device will be mandated to have secure passcodes, file-based encryption, and security features. This creates layers of protection if the device is stolen or lost.

I will allow BYOD for laptops and mobile devices but will have the company provide tablets to be used on-premise but want to ensure MDM is implemented on every device that is going to be used for company purposes. In implementing MDM, the organization will be able to send/change configurations on the device for better security and in order to implement tools if the device is lost or stolen. I want to make sure that storage segmentation is on laptops and mobile devices to ensure that personal data does not interfere with company data when organizations need to manage it or protect it in a specific way. Laptops and mobile devices will have location security on them since users will be able to travel with them. Features such as an alarm, last known location, remote lockout, and thief picture will be implemented. In addition to the location settings that laptops have, the mobile devices of high-risk users will have motion detection to track walking patterns to detect if an unauthorized person is carrying the device. Tablets will be able to connect to only specific subnets within departments where these devices are needed. Laptops and mobile devices will be able to connect to the network with certain permissions. Mobile devices will have limited access to data in comparison to laptops since there are fewer vulnerabilities on laptops.

References

Ciampa, M. (2021). *EBook: CompTIA security+ guide to network security fundamentals*.
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